



Customer Shopping Behavior Analysis

This study analyzes 3,900 purchases across multiple categories to surface spending patterns, customer segments, product preferences, subscription behavior, and actionable recommendations for revenue and retention strategies.



Project & Dataset Overview

Project goal: uncover insights from transactional data (3,900 rows, 18 columns) to guide strategic decisions. Key features include demographics (age, gender, location, subscription), purchase details (item, category, amount, season, size, color), and behavior signals (discounts, promo codes, frequency, review rating, shipping).

Missing data: 37 values in the Review Rating column. Summary stats: mean purchase amount \$59.76, average review rating 3.75, age mean 44.07.

Python — Preparation & Feature Engineering

Data loading with pandas, structural checks via `df.info()` and `.describe()`. Missing review ratings imputed using category median. Columns standardized to snake_case. Engineered features: `age_group` (binned) and `purchase_frequency_days`. Determined `promo_code_used` redundant with `discount_applied` and dropped `promo_code_used`. Cleaned DataFrame loaded into PostgreSQL for analysis.





SQL Analysis — Revenue & Customer Segments

Revenue by gender: Male \$157,890 vs Female \$75,191. Customer segmentation produced 3 groups: Loyal (3,116), Returning (701), New (83). Repeat buyers (>5 purchases): mostly non-subscribers (2,518 No vs 958 Yes).

Subscribers

Yes: 1,053 customers, avg spend \$59.49, total revenue \$62,645

Non-Subscribers

No: 2,847 customers, avg spend \$59.87, total revenue \$170,436

Top Products, Ratings & Discount Dependence

Top products by rating: Gloves (3.86), Sandals (3.84), Boots (3.82), Hat (3.80), Skirt (3.78). Discount-dependent products (highest discount rates): Hat 50.00%, Sneakers 49.66%, Coat 49.07%, Sweater 48.17%, Pants 47.37% — signalling need to review discount policy and margin impact.



Gloves

Highest avg rating (3.86).



Sandals

High rating (3.84) and strong sales.



Coat & Sweater

High discount dependence — review margins.

Purchase Behavior & Shipping Insights

Average purchase: \$59.76. Shipping comparison shows similar spend: Standard \$58.46 vs Express \$60.48 — express buyers slightly higher spend. Identified 839 rows of discount-using high spenders (customers using discounts but spending above average), suggesting targeted upsell opportunities.

Opportunity

Promote higher-margin items to discount users who still spend above average; tailor shipping offers to convert express shoppers into subscribers.





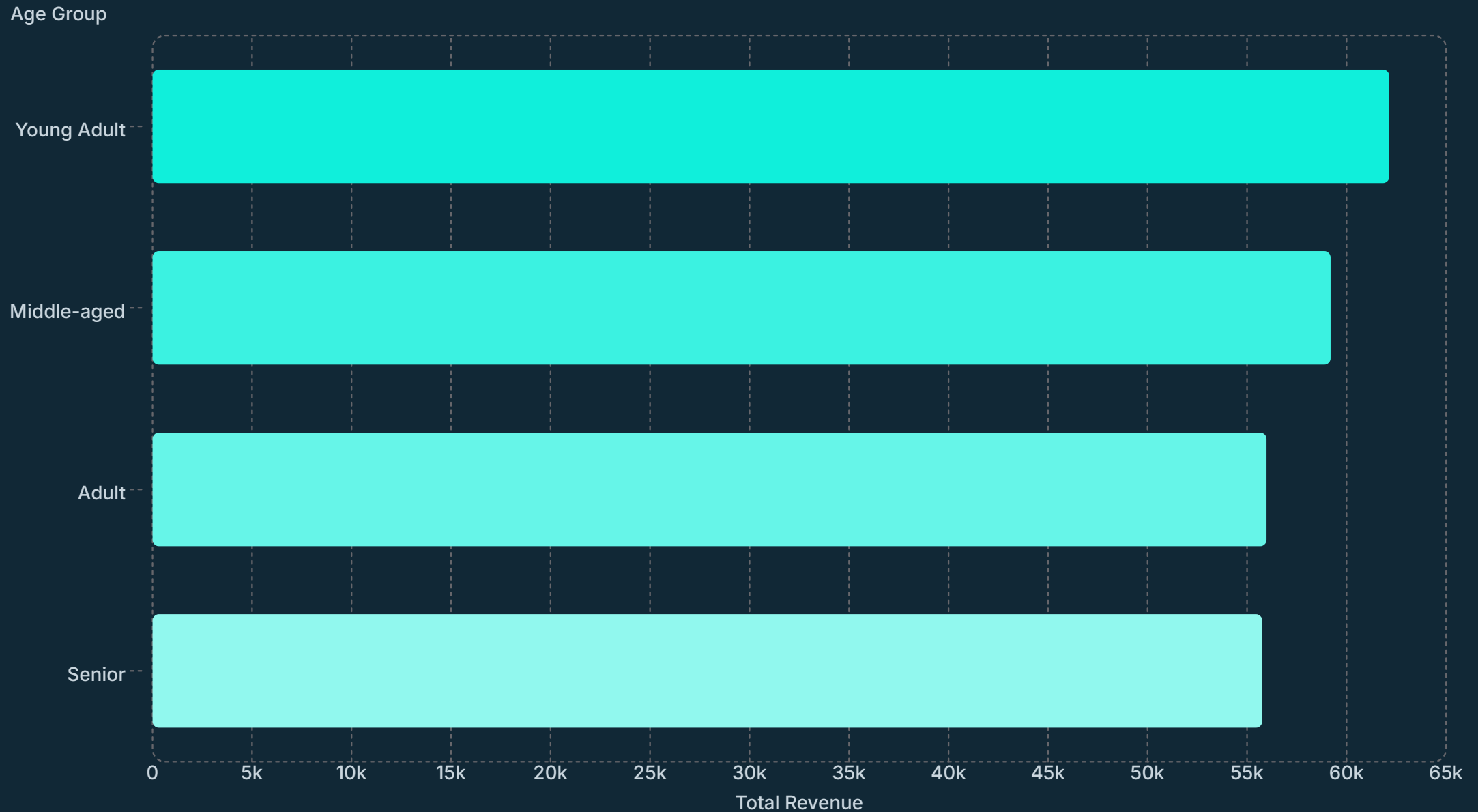
Category Performance & Top Items

Top items per category (by orders): Accessories — Jewelry (171), Sunglasses (161), Belt (161). Clothing — Blouse (171), Pants (171), Shirt (169). Footwear — Sandals (160), Shoes (150), Sneakers (145). Outerwear — Jacket (163), Coat (161).

Use product positioning to highlight best-sellers and high-rated items in marketing campaigns.

Age Group Revenue Contribution

Revenue by age group: Young Adult \$62,143; Middle-aged \$59,197; Adult \$55,978; Senior \$55,763. Young Adults are the largest revenue contributors, followed closely by Middle-aged customers — prioritize targeted marketing and product assortments for these segments.



The bar chart shows where revenue is concentrated — useful for channel and creative targeting.

Power BI Dashboard & Key Metrics

Interactive dashboard built in Power BI consolidates KPIs and filters for Subscription Status, Gender, Category, and Shipping Type. Key KPIs: 3.9K customers, \$59.76 avg purchase, 3.75 avg review rating, subscription split Yes 27% / No 73%.



The dashboard enables drilldowns by age group and category to support fast business decisions and A/B test planning.

Business Recommendations & Next Steps

Boost Subscriptions

Promote exclusive benefits and targeted offers to convert frequent buyers and express-shipping users into subscribers.

Customer Loyalty Programs

Reward repeat buyers to move them into the Loyal segment and increase lifetime value.

Review Discount Policy

Balance short-term sales with long-term margin control; target discounts to customer segments that increase basket size.

Product Positioning & Targeted Marketing

Highlight top-rated and best-selling products in campaigns; prioritize Young Adult and Middle-aged segments for personalized offers.

Next steps: implement A/B tests on subscription offers, refine discount targeting, and expand dashboard filters for cohort analysis to track impact.

